Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

SINGAPORE WORKFORCE SKILLS QUALIFICATIONS

Claude Certified Architect Foundation

TGS Ref No: TGS-2026061312

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.4

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	5 September 2026	Initial CCAR-F-aligned release	Tertiary Infotech Academy Pte Ltd
1.1	5 September 2026	OCR-derived domain integration; removed standalone screenshot gallery	Tertiary Infotech Academy Pte Ltd
1.2	5 September 2026	Dark-theme rebuild; replaced every source screenshot with editable OCR-derived models	Tertiary Infotech Academy Pte Ltd
1.3	5 September 2026	Complete reference-to-domain audit; added task-level field-pattern synthesis and traceability	Tertiary Infotech Academy Pte Ltd
2.0	5 November 2026	Aligned to trainer deck v2.0 (148 slides): every topic now alternates concept and lab. Timetable keeps the v1.3 clock times. Six simulators replace Labs 1 to 6 and run as a local website or through Claude Code. Sets out all 61 labs with their slide numbers. Moved to the house cover, record and contents format.	Tertiary Infotech Academy Pte Ltd
2.1	5 November 2026	Aligned to trainer deck v2.2 and Learner Guide v2.1. All six simulators now run in Python on the Claude Agent SDK with uv, through each learner's Claude login. No API key. Resources updated.	Tertiary Infotech Academy Pte Ltd
2.2	6 November 2026	Aligned to the Case Study: five tasks, one per learning outcome, each with one lab screenshot. Lab numbering and simulator pages updated.	Tertiary Infotech Academy Pte Ltd
2.3	6 November 2026	Each lab now shows the right answer as soon as the learner picks, then simulates that answer only, with a built-in terminal on the page. Aligned to trainer deck v2.4 and Learner Guide v2.3.	Tertiary Infotech Academy Pte Ltd
2.4	6 November 2026	Lab pages now show every step as it happens, with Claude's real stop_reason on each turn, and drop the saved-run button. Spend figures re-measured. Aligned to trainer deck v2.5 and Learner Guide v2.4.	Tertiary Infotech Academy Pte Ltd

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Course Overview

You run this course from the slides: concept, then a lab on a live simulator, then the next concept. Learners leave having run the six systems the CCAR-F exam asks about, and having been proved wrong at least once by their own evidence. Slide numbers below refer to trainer deck v2.5.

Item	Detail
Course title	Claude Certified Architect Foundation
TGS Ref No	TGS-2026061312
Duration	2 days, 16 hours: 14 hours training + 2 hours assessment
Daily timing	9:00 am to 6:00 pm, 1-hour lunch not counted
Delivery	Facilitated classroom. Each topic alternates concept slides and labs on six simulators
Assessment	Written Assessment (WA-SAQ) 60 min + Case Study (CS) 60 min, open book

Learning Outcomes

By the end of this course, participants will be able to:

- Design bounded agentic loops and multi-agent orchestration with explicit handoffs, state and stop conditions.
- Engineer reliable tool and MCP integrations with precise schemas, structured errors and least-privilege tool distribution.
- Configure Claude Code projects with scoped instructions, reusable skills, rules, plan mode and CI/CD workflows.
- Produce validated structured outputs using clear criteria, few-shot examples, batching and review loops.
- Manage context, uncertainty, provenance and human escalation in large, long-running AI systems.

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Daily Schedule

33 labs run in class. The other 28 are self-practice, and each still has its own slide and page on the simulator website. On each lab page, learners pick an answer, see at once why it is right or wrong, then watch the right answer work.

Day 1

Time	Duration	Topic / Activity	Method	Slides
9:00 to 9:30	30 min	Welcome, exam blueprint, six simulators, setup check Take the AM digital attendance. Show the LMS download. Run the setup on slide 22 and open Simulator 1, Start here, on every laptop.	Interactive presentation	1–23
9:30 to 11:30	120 min	Topic 1: Agentic Architecture & Orchestration Tasks 1.1 to 1.4. Labs 1, 3, 5, 6, 9, 10 Project Lab 1 first: pick an answer, read why, press Simulate it live. Then learners drive.	Interactive presentation, demonstration, drill and practice	24–43
11:30 to 11:45	15 min	Tea break		
11:45 to 13:00	75 min	Topic 1, continued Tasks 1.5 to 1.7. Labs 12, 15, 16 On Simulator 1, have learners switch a hook off in Play it live and watch what gets through.	Interactive presentation, demonstration, drill and practice	44–53
13:00 to 14:00	60 min	Lunch (not counted)		
14:00 to 15:30	90 min	Topic 2: Tool Design & MCP Integration Tasks 2.1 to 2.5. Labs 18, 19, 21, 23, 25 Take the PM digital attendance. Ask for one tool description written aloud before Lab 18.	Interactive presentation, demonstration, drill and practice	54–72
15:30 to 15:45	15 min	Tea break		
15:45 to 18:00	135 min	Topic 3: Claude Code Configuration & Workflows Tasks 3.1 to 3.6. Labs 27, 29, 30, 31, 33, 36 Use Play it live on Simulator 2 to show which instructions really load.	Interactive presentation, demonstration, drill and practice	73–95

Day 1 total, excluding lunch: 480 minutes (8 hours), all training.

Day 2

Time	Duration	Topic / Activity	Method	Slides
9:00 to 11:15	135 min	Topic 4: Prompt Engineering & Structured Output Tasks 4.1 to 4.6. Labs 38, 40, 43, 44, 45, 48, 50 Take the AM digital attendance. Have learners say why an answer is right before they press Simulate it live.	Interactive presentation, demonstration, drill and practice	96–119
11:15 to 12:15	60 min	Topic 5: Context Management & Reliability Tasks 5.1 to 5.2. Labs 51, 54 On Simulator 1, Play it live, have learners try to talk the agent past an escalation.	Interactive presentation, demonstration, drill and practice	120–129
12:15 to 12:30	15 min	Tea break		

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Time	Duration	Topic / Activity	Method	Slides
12:30 to 13:00	30 min	Topic 5, continued Task 5.3. Lab 55 Read the coordinator's error context in the evidence, line by line.	Interactive presentation, demonstration, drill and practice	130–131
13:00 to 14:00	60 min	Lunch (not counted)		
14:00 to 15:00	60 min	Topic 5, continued Tasks 5.4 to 5.6. Labs 56, 60, 61 Take the PM digital attendance.	Interactive presentation, demonstration, drill and practice	132–141
15:00 to 15:45	45 min	Recap, practice exam and assessment briefing Explain the Case Study: five tasks, one per learning outcome, each needing one screenshot from a lab the learner runs. Learners complete TRAQOM and the Assessment digital attendance.	Interactive presentation	142–147, 11–13
15:45 to 16:00	15 min	Tea break		
16:00 to 17:00	60 min	Written Assessment (WA-SAQ): 10 questions, open book	Assessment	
17:00 to 18:00	60 min	Case Study (CS): 5 tasks on the Northstar Services scenario, one lab screenshot per task, open book	Assessment	

Day 2 total, excluding lunch: 480 minutes (8 hours): 6 hours training + 2 hours assessment.

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Topic 1: Agentic Architecture & Orchestration · Slides 24–53

27% of the exam. 7 task statements, 17 labs, 9 of them in class. Divider on slide 24, topic map on slide 25. Your job here: make learners feel the difference between asking and enforcing. Lab 9 and Lab 12 do it fastest.

Task 1.1 · Slides 26–29

The loop runs on stop_reason, and nothing else Design and implement agentic loops for autonomous task execution.

Lab 1 · Slide 28 · Watch one refund go round the loop

In class on Simulator 1. Built from exam guide task statement 1.1 (no practice question covers this angle).

Lab 2 · Slide 29 · Claude re-asks because the tool_result never came back

Self-practice on Simulator 2. SkillCertPro practice exam Part 1, Q17 (unofficial third-party question).

Task 1.2 · Slides 30–32

One hub. Every message goes through it. Orchestrate multi-agent systems with coordinator-subagent patterns.

Lab 3 · Slide 31 · Every subagent succeeds, and the report still misses half the topic

In class on Simulator 3. Official exam guide, sample question 7.

Lab 4 · Slide 32 · Same brief to every subagent, same findings four times

Self-practice on Simulator 6. SkillCertPro practice exam Part 1, Q36 (unofficial third-party question).

Task 1.3 · Slides 33–38

The Task tool is how a coordinator spawns anything Configure subagent invocation, context passing, and spawning.

Lab 5 · Slide 35 · A /parallel-review that can't fan out because Task isn't allowed

In class on Simulator 2. SkillCertPro practice exam Part 1, Q25 (unofficial third-party question).

Lab 6 · Slide 36 · Same reviewers, same findings, and the PR comment loses them all

In class on Simulator 5. SkillCertPro practice exam Part 1, Q2 (unofficial third-party question).

Lab 7 · Slide 37 · Polished report, weak citations: route information, not just tasks

Self-practice on Simulator 3. SkillCertPro practice exam Part 1, Q10 (unofficial third-party question).

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Lab 8 · Slide 38 · Later subagents re-run searches the first ones already finished

Self-practice on Simulator 3. SkillCertPro practice exam Part 1, Q35 (unofficial third-party question).

Task 1.4 · Slides 39–43

Identity before refund is code, not a polite request Implement multi-step workflows with enforcement and handoff patterns.

Lab 9 · Slide 41 · 12% of cases skip get_customer: fix it with a gate, not a sentence

In class on Simulator 1. Official exam guide, sample question 1.

Lab 10 · Slide 42 · Approval must never post before the passing test artifact exists

In class on Simulator 5. SkillCertPro practice exam Part 1, Q3 (unofficial third-party question).

Lab 11 · Slide 43 · Claude guesses a field and submits without validating

Self-practice on Simulator 6. SkillCertPro practice exam Part 1, Q12 (unofficial third-party question).

Task 1.5 · Slides 44–48

PreToolUse decides. PostToolUse reshapes. Apply Agent SDK hooks for tool call interception and data normalization.

Lab 12 · Slide 46 · The prompt forgot the $500 rule. The hook did not.

In class on Simulator 1. Built from exam guide task statement 1.5 (no practice question covers this angle).

Lab 13 · Slide 47 · Shrink noisy npm test output before Claude reads it

Self-practice on Simulator 2. SkillCertPro practice exam Part 1, Q9 (unofficial third-party question).

Lab 14 · Slide 48 · rm -rf meets a hook, not a polite request

Self-practice on Simulator 4. Built from exam guide task statement 1.5 (no practice question covers this angle).

Task 1.6 · Slides 49–50

Known steps get a chain. Unknown ground gets a plan that adapts. Design task decomposition strategies for complex workflows.

Lab 15 · Slide 50 · 60-page packets: section passes first, reconcile second

In class on Simulator 6. SkillCertPro practice exam Part 1, Q31 (unofficial third-party question).

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Task 1.7 · Slides 51–53

Resume when the facts still hold. Start fresh when they don't. Manage session state, resumption, and forking.

Lab 16 · Slide 52 · Two attempts, one analysis: fork the session, checkpoint the files

In class on Simulator 2. SkillCertPro practice exam Part 1, Q5 (unofficial third-party question).

Lab 17 · Slide 53 · Resume, and say exactly which file changed

Self-practice on Simulator 4. Built from exam guide task statement 1.7 (no practice question covers this angle).

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Topic 2: Tool Design & MCP Integration · Slides 54–72

18% of the exam. 5 task statements, 9 labs, 5 of them in class. Divider on slide 54, topic map on slide 55. Your job here: keep bringing it back to the agent's next move. Every description and error exists so Claude can choose well.

Task 2.1 · Slides 56–57

The description is the router Design effective tool interfaces with clear descriptions and boundaries.

Lab 18 · Slide 57 · One analyze_document tool does three jobs badly

In class on Simulator 6. SkillCertPro practice exam Part 1, Q11 (unofficial third-party question).

Task 2.2 · Slides 58–61

"Operation failed" tells the agent nothing it can use Implement structured error responses for MCP tools.

Lab 19 · Slide 60 · The refund failed. The customer was told it worked.

In class on Simulator 1. SkillCertPro practice exam Part 1, Q29 (unofficial third-party question).

Lab 20 · Slide 61 · Retry the timeout once, then answer from the real record

Self-practice on Simulator 1. Built from exam guide task statement 2.2 (no practice question covers this angle).

Task 2.3 · Slides 62–65

18 tools degrade selection. Give each agent 4 or 5. Distribute tools appropriately across agents and configure tool choice.

Lab 21 · Slide 64 · Review and test generation fight over one Claude call

In class on Simulator 5. SkillCertPro practice exam Part 1, Q15 (unofficial third-party question).

Lab 22 · Slide 65 · Give synthesis one narrow tool, not the whole web

Self-practice on Simulator 3. Official exam guide, sample question 9.

Task 2.4 · Slides 66–68

Shared servers live in the repo. Secrets never do. Integrate MCP servers into Claude Code and agent workflows.

Lab 23 · Slide 67 · Commit the server, never the token

In class on Simulator 2. Built from exam guide task statement 2.4 (no practice question covers this angle).

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Lab 24 · Slide 68 · Read the catalogue before opening a single file

Self-practice on Simulator 4. Built from exam guide task statement 2.4 (no practice question covers this angle).

Task 2.5 · Slides 69–72

Search narrows. Read confirms. Edit changes. Select and apply built-in tools (Read, Write, Edit, Bash, Grep, Glob) effectively.

Lab 25 · Slide 71 · /find-callers needs Grep with a glob, not Bash

In class on Simulator 4. SkillCertPro practice exam Part 1, Q37 (unofficial third-party question).

Lab 26 · Slide 72 · Edit says the anchor isn't unique. Now what?

Self-practice on Simulator 4. Built from exam guide task statement 2.5 (no practice question covers this angle).

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Topic 3: Claude Code Configuration & Workflows · Slides 73–95

20% of the exam. 6 task statements, 11 labs, 6 of them in class. Divider on slide 73, topic map on slide 74. Your job here: show that configuration is a team sport. What sits in one person's home folder does not exist for anyone else.

Task 3.1 · Slides 75–77

If it's in ~/.claude, your teammate never sees it Configure CLAUDE.md files with appropriate hierarchy, scoping, and modular organization.

Lab 27 · Slide 77 · Which instructions actually reached Claude?

In class on Simulator 2. Built from exam guide task statement 3.1 (no practice question covers this angle).

Task 3.2 · Slides 78–80

Skills run on demand. CLAUDE.md is always on. Create and configure custom slash commands and skills.

Lab 28 · Slide 79 · A /review command the whole team gets through git

Self-practice on Simulator 2. Official exam guide, sample question 4.

Lab 29 · Slide 80 · Keep a noisy case review out of your main session

In class on Simulator 1. SkillCertPro practice exam Part 1, Q16 (unofficial third-party question).

Task 3.3 · Slides 81–83

Test files live everywhere, so scope the rule by glob Apply path-specific rules for conditional convention loading.

Lab 30 · Slide 83 · Testing rules for test files spread across every folder

In class on Simulator 3. SkillCertPro practice exam Part 1, Q7 (unofficial third-party question).

Task 3.4 · Slides 84–88

Plan mode for blast radius. Direct for a known fix. Determine when to use plan mode vs direct execution.

Lab 31 · Slide 86 · Replacing the auth layer: nobody agrees on the boundary, so plan

In class on Simulator 2. SkillCertPro practice exam Part 1, Q8 (unofficial third-party question).

Lab 32 · Slide 87 · Plan first, then approve into an editing mode

Self-practice on Simulator 2. SkillCertPro practice exam Part 1, Q30 (unofficial third-party question).

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Lab 33 · Slide 88 · One guard clause doesn't need a plan

In class on Simulator 2. SkillCertPro practice exam Part 1, Q34 (unofficial third-party question).

Task 3.5 · Slides 89–91

Show the output you want. Prose gets interpreted. Apply iterative refinement techniques for progressive improvement.

Lab 34 · Slide 90 · Unknown requirements? Let Claude interview you first

Self-practice on Simulator 5. SkillCertPro practice exam Part 1, Q14 (unofficial third-party question).

Lab 35 · Slide 91 · Two linked defects go in one detailed request

Self-practice on Simulator 3. SkillCertPro practice exam Part 1, Q27 (unofficial third-party question).

Task 3.6 · Slides 92–95

CI has no keyboard, so run claude -p Integrate Claude Code into CI/CD pipelines.

Lab 36 · Slide 94 · The nightly job hangs because nobody can press Enter

In class on Simulator 5. SkillCertPro practice exam Part 1, Q18 (unofficial third-party question).

Lab 37 · Slide 95 · A shared slash command still needs -p in CI

Self-practice on Simulator 5. SkillCertPro practice exam Part 1, Q38 (unofficial third-party question).

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Topic 4: Prompt Engineering & Structured Output · Slides 96–119

20% of the exam. 6 task statements, 13 labs, 7 of them in class. Divider on slide 96, topic map on slide 97. Your job here: separate shape from meaning. A valid record can still be a wrong one, and Lab 43 proves it.

Task 4.1 · Slides 98–100

"Be conservative" is not a criterion Design prompts with explicit criteria to improve precision and reduce false positives.

Lab 38 · Slide 99 · "Only high confidence" didn't quiet the reviewer

In class on Simulator 5. SkillCertPro practice exam Part 1, Q39 (unofficial third-party question).

Lab 39 · Slide 100 · Switch off the one noisy category, keep the accurate ones

Self-practice on Simulator 3. SkillCertPro practice exam Part 1, Q20 (unofficial third-party question).

Task 4.2 · Slides 101–102

Two to four examples draw the line prose can't Apply few-shot prompting to improve output consistency and quality.

Lab 40 · Slide 102 · "Clean this up" means three things: show which one you mean

In class on Simulator 2. SkillCertPro practice exam Part 1, Q23 (unofficial third-party question).

Task 4.3 · Slides 103–106

The schema guarantees shape, never meaning Enforce structured output using tool use and JSON schemas.

Lab 41 · Slide 105 · Vague passages forced into a category they don't fit

Self-practice on Simulator 3. SkillCertPro practice exam Part 1, Q28 (unofficial third-party question).

Lab 42 · Slide 106 · Know your tool_choice options, and check what came back

Self-practice on Simulator 6. Built from exam guide task statement 4.3 (no practice question covers this angle).

Task 4.4 · Slides 107–109

Retry with the error, or stop if the data isn't there Implement validation, retry, and feedback loops for extraction quality.

Lab 43 · Slide 108 · Valid JSON, wrong refund: add business rules

In class on Simulator 1. SkillCertPro practice exam Part 1, Q4 (unofficial third-party question).

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Lab 44 · Slide 109 · Retry when the format is wrong, stop when the data is absent

In class on Simulator 6. Built from exam guide task statement 4.4 (no practice question covers this angle).

Task 4.5 · Slides 110–114

Half the price, up to 24 hours, no tool loop Design efficient batch processing strategies.

Lab 45 · Slide 112 · Prove the template before 35,000 transcripts

In class on Simulator 1. SkillCertPro practice exam Part 1, Q1 (unofficial third-party question).

Lab 46 · Slide 113 · Client tools can't run inside a batch

Self-practice on Simulator 1. SkillCertPro practice exam Part 1, Q26 (unofficial third-party question).

Lab 47 · Slide 114 · Nightly job on Batches, upload flow stays synchronous

Self-practice on Simulator 6. SkillCertPro practice exam Part 1, Q32 (unofficial third-party question).

Task 4.6 · Slides 115–119

The author is the worst reviewer of their own draft Design multi-instance and multi-pass review architectures.

Lab 48 · Slide 117 · The author reviewing its own draft misses its own errors

In class on Simulator 3. SkillCertPro practice exam Part 1, Q6 (unofficial third-party question).

Lab 49 · Slide 118 · Source-level passes first, integration second

Self-practice on Simulator 3. SkillCertPro practice exam Part 1, Q19 (unofficial third-party question).

Lab 50 · Slide 119 · 14 files, shallow comments: split the review

In class on Simulator 5. Official exam guide, sample question 12.

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Topic 5: Context Management & Reliability · Slides 120–141

15% of the exam. 6 task statements, 11 labs, 6 of them in class. Divider on slide 120, topic map on slide 121. Your job here: make failure visible. A blank result pretending to be success is the most expensive bug in the course.

Task 5.1 · Slides 122–126

Facts live outside the summary Manage conversation context to preserve critical information across long interactions.

Lab 51 · Slide 124 · Refund amount jumped to the wrong order

In class on Simulator 1. SkillCertPro practice exam Part 1, Q40 (unofficial third-party question).

Lab 52 · Slide 125 · 40-field tool results push the review to its limit

Self-practice on Simulator 5. SkillCertPro practice exam Part 1, Q13 (unofficial third-party question).

Lab 53 · Slide 126 · Taxonomy drift: write it to a state file

Self-practice on Simulator 3. SkillCertPro practice exam Part 1, Q21 (unofficial third-party question).

Task 5.2 · Slides 127–129

Escalate on triggers, not on tone Design effective escalation and ambiguity resolution patterns.

Lab 54 · Slide 129 · 55% first-contact resolution: calibrate with criteria and examples

In class on Simulator 1. Official exam guide, sample question 3.

Task 5.3 · Slides 130–131

Say what failed, what you tried, what you still have Implement error propagation strategies across multi-agent systems.

Lab 55 · Slide 131 · The search timed out: tell the coordinator everything

In class on Simulator 3. Official exam guide, sample question 8.

Task 5.4 · Slides 132–135

Delegate the noise, keep the signal Manage context effectively in large codebase exploration.

Lab 56 · Slide 133 · Delegate the noisy search, keep a scratchpad

In class on Simulator 4. Built from exam guide task statement 5.4 (no practice question covers this angle).

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Lab 57 · Slide 134 · Keep discovery out of the main thread

Self-practice on Simulator 6. SkillCertPro practice exam Part 1, Q22 (unofficial third-party question).

Lab 58 · Slide 135 · Crash recovery from a manifest

Self-practice on Simulator 4. Built from exam guide task statement 5.4 (no practice question covers this angle).

Task 5.5 · Slides 136–138

97% overall can hide the one field that fails Design human review workflows and confidence calibration.

Lab 59 · Slide 137 · Acceptance looks great. Two fields don't.

Self-practice on Simulator 2. SkillCertPro practice exam Part 1, Q33 (unofficial third-party question).

Lab 60 · Slide 138 · The aggregate masks a segment: break it and watch

In class on Simulator 6. Built from exam guide task statement 5.5 (no practice question covers this angle).

Task 5.6 · Slides 139–141

The source travels with the claim, all the way to the report Preserve information provenance and handle uncertainty in multi-source synthesis.

Lab 61 · Slide 141 · A vendor benchmark isn't a peer-reviewed study

In class on Simulator 3. SkillCertPro practice exam Part 1, Q24 (unofficial third-party question).

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Resources Required

- Trainer deck v2.5 (148 slides, PPT) with speaker notes, and learner slides (PDF). The notes that begin SAY THIS ALOUD cover ideas the slides leave out, so say them.
- Learner Guide v2.4, and the `labs` folder: six Python simulators on the Claude Agent SDK, started with `uv run serve.py`, and 61 lab guides.
- A laptop per learner with `uv` and Claude Code signed in (`claude auth login`). Internet access is required, because live runs call Claude through each learner's login. No API key is needed.
- A budget for live runs. On `claude-sonnet-5-5`, the default, Lab 1 costs one to three US cents and the long research run in Lab 3 about 40 cents. `claude-haiku-5-5` spends least.
- TRAQOM digital attendance QR codes (AM, PM, Assessment), whiteboard and markers.
- WA-SAQ and CS papers, Assessment Summary Record and Appeal Form.

Assessment

Two papers on Day 2, each one hour, open book: Written Assessment at 4:00 pm and Case Study at 5:00 pm. Candidates need at least 75% attendance to be eligible. Courseware and answer submission are on the LMS: https://lms-tms.tertiaryinfotech.com/

Run the flow in this order: TRAQOM survey, Assessment digital attendance, WA-SAQ and Case Study, submit answers on the LMS, sign the Assessment Summary Record (slides 13 and 146).

Each Case Study task names one lab and asks for one screenshot from the learner's own live run, then a written answer. The five labs took about three minutes in total on `claude-sonnet-5-5`, so runs fit inside the hour. Say this at the briefing (slides 12 and 145 carry the same line).

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Where each paper item is taught

Paper item	Where it is taught	Slides	Labs
Question 1 (K1)	Tasks 1.1, 1.4	26–29, 39–43	1, 2, 9, 10, 11
Question 2 (K2)	Tasks 1.2, 1.3, 1.6	30–32, 33–38, 49–50	3, 4, 5, 6, 7, 8, 15
Question 3 (K3)	Tasks 1.5, 1.7	44–48, 51–53	12, 13, 14, 16, 17
Question 4 (K4)	Tasks 2.1, 2.2	56–57, 58–61	18, 19, 20
Question 5 (K5)	Tasks 2.3, 2.4, 2.5	62–65, 66–68, 69–72	21, 22, 23, 24, 25, 26
Question 6 (K6)	Tasks 3.1, 3.2, 3.3	75–77, 78–80, 81–83	27, 28, 29, 30
Question 7 (K7)	Tasks 3.4, 3.5, 3.6	84–88, 89–91, 92–95	31, 32, 33, 34, 35, 36, 37
Question 8 (K8)	Tasks 4.1, 4.2, 4.3, 4.4	98–100, 101–102, 103–106, 107–109	38, 39, 40, 41, 42, 43, 44
Question 9 (K9)	Tasks 4.5, 4.6	110–114, 115–119	45, 46, 47, 48, 49, 50
Question 10 (K10)	Tasks 5.1, 5.2, 5.3, 5.4, 5.5, 5.6	122–126, 127–129, 130–131, 132–135, 136–138, 139–141	51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61

Case Study task	Domain	Simulator	Slide	Lab
Task 1 (A1, LO1)	Domain 1	Simulator 1	41	Lab 9
Task 2 (A2, LO2)	Domain 2	Simulator 1	60	Lab 19
Task 3 (A3, LO3)	Domain 3	Simulator 2	77	Lab 27
Task 4 (A4, LO4)	Domain 4	Simulator 6	109	Lab 44
Task 5 (A5, LO5)	Domain 5	Simulator 4	133	Lab 56

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